

CISD 43 – BIG DATA (Spring, 2026) Final Project
Khim Bravo Chea

Project Name: US Housing Predictive and Analysis

Project Explanation and Conclusion:

For this final project, the primary objective was to identify the key factors that accurately predict real estate prices using a linear regression model applied to the USA Housing dataset. Utilizing Python libraries such as pandas, seaborn, and scikit-learn, the data was explored and visualized to understand how various features, such as average area income, house age, room counts, and population impact the final property price. The subsequent evaluation metrics demonstrate that this predictive model is exceptionally robust and reliable. It achieved a remarkably high correlation of 0.958, indicating that the model's estimates scale in near-perfect synchronization with actual true market values. Furthermore, while the Root Mean Squared Error (RMSE) measured roughly \$101,108, the average relative error was just 7.62%. Ultimately, the analysis confirms that linear regression is a highly dependable tool for real estate valuation in this context, consistently predicting housing prices within a very narrow and acceptable margin of error.